

```
%oldchk=/groups/rmaier/Ravindu/PbRhaC14C146C/PbRhaC14C14T2.chk
%nprocshared=64
%mem=800GB
%chk=PbRhaC14C14T3.chk
# freq apfd/gen scrf=(solvent=water) guess=read pseudo=read scf=xqc
```

PbRhaC14C14T2

0 1

Pb	-0.27972900	0.89958000	-1.09320900
O	1.40924700	-0.66221900	1.64735100
O	0.63388800	1.55412400	-3.25256000
O	-0.39373400	-1.47810200	-0.06548300
C	1.71953400	-4.47232700	0.72081500
O	2.91935100	-3.72015000	0.48181600
C	2.97013000	-2.48899800	1.17204200
C	1.91257400	-1.50628500	0.61529400
C	0.76416500	-2.30595700	0.01255800
C	0.50430300	-3.54602000	0.85259000
O	1.46277400	-0.23619700	-2.26507500
C	1.50367100	0.64159900	-3.17964800
C	2.66584400	0.61607200	-4.14077300
C	3.82663700	1.46937000	-3.60856100
O	4.10616700	1.12509000	-2.23091300
C	3.55391100	2.95850500	-3.62271600
C	4.74952900	3.79077100	-3.18083700
C	4.46067700	5.28548900	-3.20830200
C	5.64458100	6.13154200	-2.75990700
C	5.35768200	7.62695100	-2.78317800
C	4.72364100	-0.03508400	-1.99636500
O	5.10888100	-0.78460500	-2.86540200
C	4.86857300	-0.28740600	-0.51850000
C	4.80175500	-1.78751300	-0.20149900
O	4.26819400	-1.99019400	1.11914600
C	1.86817800	-5.39023200	1.91838900
O	-0.67158400	-4.21464500	0.43692100
C	6.17058200	-2.44016500	-0.27263800
C	6.12930500	-3.95647500	-0.17941400
C	7.51427600	-4.58787500	-0.20107400
C	7.47501500	-6.10767300	-0.11335800
C	8.85499700	-6.75127400	-0.13199800
O	-0.88368800	-0.26657300	4.43466400
O	-2.85713300	1.33606000	-1.99708800
O	0.82974700	-2.46800000	3.90023100
C	-2.05749900	-4.12827900	3.90253600
O	-1.92813800	-3.30694600	2.74041200
C	-1.89599200	-1.91983600	3.00045100
C	-1.38165500	-1.59056200	4.39998000
C	-0.27059300	-2.53857500	4.79036900
C	-0.82159000	-3.98039300	4.79931300

O	-1.67532700	-0.51851000	-2.26756400
C	-2.75986900	0.15583000	-2.37771600
C	-3.94345600	-0.56063200	-2.99428000
C	-5.22169800	-0.33265300	-2.21343400
O	-4.96423300	-0.83554700	-0.87484300
C	-6.40208300	-1.07109600	-2.81795100
C	-7.71017500	-0.88566300	-2.06286100
C	-8.86886200	-1.62532400	-2.71783300
C	-10.18147600	-1.46238800	-1.96336500
C	-11.34352800	-2.20376300	-2.61062400
C	-5.38192700	-0.11398500	0.17123800
O	-5.98293300	0.93162100	0.09125300
C	-4.96763700	-0.77304900	1.46261900
C	-3.44888300	-0.81991300	1.58744700
O	-3.16395700	-1.33965500	2.88976300
C	-2.26765500	-5.54802900	3.43817000
O	-1.24919100	-4.38040100	6.08788700
C	-2.78700400	0.54067400	1.40571400
C	-3.26372300	1.62884100	2.35798400
C	-2.37655100	2.86441100	2.28962000
C	-2.82349900	3.97934000	3.22396000
C	-1.93109400	5.21105700	3.15253100
H	-3.06096300	-1.51566800	0.83601400
H	-5.42956700	0.73467400	-2.12878400
H	4.14211200	-2.27547000	-0.92432700
H	4.71845400	1.23964400	-4.19802700
H	1.58168200	-5.06102400	-0.19167500
H	2.76440600	-2.63710400	2.23875900
H	2.34775400	-0.89325600	-0.17522700
H	1.05916200	-2.60204000	-0.99954800
H	0.39115800	-3.24706600	1.89621600
H	2.99971200	-0.41349200	-4.26608600
H	2.36047700	1.01821300	-5.10808300
H	3.26617200	3.23386200	-4.64330900
H	2.68611400	3.17157500	-2.98940500
H	5.04730100	3.49536500	-2.16871100
H	5.60732200	3.57081800	-3.82971500
H	4.16360400	5.58219500	-4.22290300
H	3.59627900	5.49992100	-2.56600900
H	5.94038600	5.83342800	-1.74523700
H	6.50946200	5.91653200	-3.40149700
H	4.49047500	7.84061100	-2.14413000
H	5.06430500	7.92528500	-3.79853100
H	5.82908600	0.13123100	-0.19637500
H	4.09703700	0.26700800	0.01478400
H	0.95334300	-5.97344100	2.05015300
H	2.03626700	-4.81404500	2.83162100
H	2.70075200	-6.08248800	1.77720400
H	-1.32883000	-4.02486000	1.12594300
H	6.63693000	-2.13207500	-1.21439200

H	6.78685600	-2.02496400	0.53580300
H	5.60140800	-4.24955800	0.73366900
H	5.53303800	-4.35304700	-1.01093200
H	8.04111500	-4.28917500	-1.11733200
H	8.10982000	-4.19062700	0.63186900
H	6.94752400	-6.40437600	0.80305900
H	6.87786700	-6.50386500	-0.94557000
H	9.45213700	-6.35609200	0.70071500
H	9.38261000	-6.45460300	-1.04837400
H	-2.93094600	-3.79502100	4.47740000
H	-1.21771400	-1.50018300	2.25327200
H	-2.20215400	-1.71097600	5.11898800
H	0.06005100	-2.29248300	5.80814300
H	-0.02228400	-4.63438900	4.42992600
H	-4.07868500	-0.17250400	-4.00910400
H	-3.72419100	-1.62704100	-3.07301100
H	-6.51568900	-0.71534300	-3.84865700
H	-6.15551400	-2.13806400	-2.88271500
H	-7.59529100	-1.24101600	-1.03227700
H	-7.94569200	0.18265500	-1.99262100
H	-8.99379000	-1.26936100	-3.74899000
H	-8.62298300	-2.69277900	-2.79443300
H	-10.05354800	-1.81642100	-0.93175000
H	-10.42729900	-0.39486000	-1.88733200
H	-11.09573700	-3.27084400	-2.68814400
H	-11.47257000	-1.84879200	-3.64182100
H	-5.39853700	-0.20878900	2.28967900
H	-5.36270000	-1.79173800	1.49101800
H	-2.32770600	-6.21323100	4.30139500
H	-1.43320800	-5.87206600	2.81100100
H	-3.19223400	-5.63771900	2.86433900
H	-0.47429800	-4.51442200	6.64278600
H	-1.70635800	0.40124800	1.53177700
H	-2.94390800	0.87027300	0.37256200
H	-4.29570900	1.90781500	2.11950300
H	-3.29399400	1.24577900	3.38668300
H	-1.34080600	2.58440300	2.52501600
H	-2.35758900	3.23771200	1.25728900
H	-3.85684200	4.26261900	2.98448600
H	-2.84433300	3.60285000	4.25533300
H	-1.90956400	5.58438000	2.11999500
H	-0.89789200	4.92556400	3.39135000
H	-1.59531500	0.32668200	4.16879000
H	0.89271300	-1.57841200	3.52229400
H	-1.02873500	-1.92136200	-0.64496100
H	2.09347000	-0.02270100	1.87210600
C	-12.65561100	-2.04451300	-1.85410900
H	-12.90416700	-0.97753500	-1.77761700
H	-12.52552700	-2.39802800	-0.82250600
C	-13.81754900	-2.78824200	-2.49913300

H	-13.94812100	-2.43451300	-3.53059700
H	-13.56842400	-3.85507500	-2.57602900
C	-15.12940900	-2.63021200	-1.74185300
H	-15.37898800	-1.56342800	-1.66557100
H	-14.99835700	-2.98322800	-0.71017500
C	-16.29105500	-3.37510500	-2.38582900
H	-16.42325400	-3.02236500	-3.41757400
H	-16.04216600	-4.44215400	-2.46244100
C	-2.37357900	6.33075900	4.08484900
H	-3.40704600	6.61540600	3.84629300
H	-2.39466300	5.95734300	5.11739600
C	-1.48125000	7.56267800	4.01181200
H	-1.46027900	7.93557100	2.97905400
H	-0.44772900	7.27773700	4.25003400
C	-1.92293600	8.68332500	4.94368600
H	-2.95666200	8.96776400	4.70574500
H	-1.94341300	8.31053900	5.97651800
C	-1.03114200	9.91548500	4.86983800
H	-1.01053500	10.28915300	3.83719900
H	0.00298900	9.63187800	5.10764200
C	6.53901100	8.47429900	-2.32969600
H	7.40649400	8.26047600	-2.96831200
H	6.83186300	8.17576700	-1.31420400
C	6.25207000	9.96978500	-2.35260800
H	5.96113400	10.26870100	-3.36852600
H	5.38321300	10.18309100	-1.71566400
C	7.43210900	10.81734100	-1.89592300
H	8.30130500	10.60340500	-2.53222000
H	7.72227800	10.51887200	-0.87962200
C	7.14556400	12.31278100	-1.91978100
H	6.85664700	12.61242800	-2.93621000
H	6.27577000	12.52751600	-1.28436100
C	8.81095200	-8.27103700	-0.04471400
H	8.28238100	-8.56721000	0.87128600
H	8.21365700	-8.66557700	-0.87761400
C	10.18965500	-8.91771000	-0.06257100
H	10.71858600	-8.62134700	-0.97828000
H	10.78691400	-8.52421900	0.77082800
C	10.14319400	-10.43751400	0.02352300
H	9.61445500	-10.73365400	0.93944000
H	9.54522300	-10.83055400	-0.80960700
C	11.52104500	-11.08574500	0.00482800
H	12.05043400	-10.79067200	-0.91118600
H	12.11991400	-10.69349900	0.83783700
C	-17.60373700	-3.21803200	-1.62846100
H	-17.85099100	-2.15163400	-1.55282800
H	-17.46977100	-3.57040200	-0.59794800
C	-18.75694300	-3.96840000	-2.28294300
H	-19.69032900	-3.84504000	-1.72652900
H	-18.92916300	-3.61175400	-3.30346000

H	-18.54493400	-5.04072000	-2.34207600
C	11.47434100	-12.60618100	0.09114800
H	10.94513500	-12.89919400	1.00665100
H	10.87539200	-12.99627600	-0.74140600
C	12.85875300	-13.24203700	0.07120400
H	13.46523400	-12.88975500	0.91167900
H	12.80638800	-14.33259800	0.13408200
H	13.39479000	-12.98788900	-0.84879300
C	8.32500000	13.16138800	-1.46130100
H	9.19322500	12.94528900	-2.09655300
H	8.61208600	12.86093900	-0.44573400
C	8.02505100	14.65486000	-1.49053700
H	8.88148300	15.24826800	-1.15826100
H	7.76543700	14.98393100	-2.50178400
H	7.17950700	14.89876700	-0.83947400
C	-1.47199400	11.03723500	5.80189500
H	-2.50528700	11.31907000	5.56332000
H	-1.49202800	10.66236400	6.83297800
C	-0.57171000	12.26346900	5.71765100
H	-0.56001300	12.67351800	4.70274500
H	-0.90237200	13.05792600	6.39257100
H	0.46057400	12.01212700	5.98133600

C H O 0

6-311+g(2d,p)

Pb 0

S	4	1.00	
	591.61124370		0.22126521076D-03
	46.757232559		0.56961959130D-02
	20.746462696		-0.21374063831
	14.610796419		0.40502620616
S	2	1.00	
	20.222637612		-0.83541883299D-01
	6.4767324865		0.97910892388
S	1	1.00	
	1.6600600927		1.0000000
S	1	1.00	
	0.80431655001		1.0000000
S	1	1.00	
	0.22627039020		1.0000000
S	1	1.00	
	0.84014530665D-01		1.0000000
P	3	1.00	
	15.189102118		0.61952303583
	14.693144415		-0.72498497086
	6.8705890048		0.37680007984
P	3	1.00	
	2.2021426123		0.40196284806
	1.2209125119		0.46058131862

		0.63367559815	0.19367655397
P	1	1.00	
		0.28202837058	1.0000000
P	1	1.00	
		0.11333375666	1.0000000
P	1	1.00	
		0.43948707430D-01	1.0000000
D	6	1.00	
		61.315369628	0.33870800787D-03
		12.372195840	0.13788683942D-01
		6.9254944983	-0.75979608103D-01
		2.3319539939	0.28113784298
		1.2108730003	0.44474512269
		0.60090478506	0.35326874351
D	1	1.00	
		0.28135869813	1.0000000
D	1	1.00	
		0.11500000000	1.0000000
F	1	1.00	
		1.0000000	1.0000000
F	1	1.00	
		0.2896200	1.0000000

PB 0

PB-ECP 3 60

f potential

2

2	3.88751200	12.20989200
2	3.81196300	16.19029100

s-f potential

4

2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

p-f potential

6

2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

d-f potential

6

2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600
2	4.56378900	14.83373100

2	3.88751200	-12.20989200
2	3.81196300	-16.19029100